



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTING TECHNOLOGIES
21CSP401L – MAJOR PROJECT

Hybrid Deep Learning and Remote Sensing for Automated Water Quality Monitoring Advancing SDG 6: Clean Water and Sanitation

Batch ID: 184

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Abstract

Ensuring timely and accurate water quality monitoring is a major challenge for achieving SDG 6 (Clean Water and Sanitation). Traditional sampling methods are resource-intensive and limited in spatial/temporal coverage. This project proposes a hybrid deep learning framework that leverages multispectral satellite imagery and advanced machine learning, combining Random Forest, Convolutional Neural Networks (CNN), and CNN-LSTM models, further integrated with explainable AI for robust, interpretable, and transferable water quality predictions. The goal is to provide an automated, scalable, and actionable decision-support tool for regional and national water resource management agencies, improving on the limitations of current ML-based approaches identified in recent literature.

Problem Statement

Water quality monitoring is critical for detecting pollution and safeguarding public health in line with SDG 6 targets. However, current methods relying on in-situ data collection are expensive, infrequent, and cannot provide comprehensive spatial coverage. Recent advances using machine learning on remote sensing data are promising, but most existing models lack transferability across regions, do not robustly handle varying conditions (sensor type, season, or pollution type), and seldom provide interpretable predictions for stakeholders. There is a need for a practical, scalable ML pipeline that combines deep learning, traditional ML, and explainability to deliver accurate and understandable water quality assessments across diverse geographies.



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Research Objectives

- O1. To design an automated water quality monitoring framework using multispectral satellite imagery for large-scale river pollution assessment.
- O2. To integrate CPCB physicochemical water quality data as ground truth for reliable labeling and validation of satellite-based predictions.
- O3. To develop a machine learning–based classification model (Random Forest) for identifying polluted (Poor) and moderately polluted water bodies.
- O4. To incorporate explainable AI techniques (SHAP) for interpreting model predictions and improving trust among regulatory and decision-making authorities.
- O5. To evaluate the scalability and generalization of the proposed framework across multiple regions and water bodies using national-level datasets.

Literature Survey

Sr. No.	Paper / Short Title (year)	Year	Key limitations reported	How we will overcome them
1.	Comparative analysis of ML models for detecting water pollution hotspots using EO data (Nature)	2025	Limited generalization across basins, insufficient use of hybrid methods, and lack of explainability.	Integrate hybrid RF + CNN/CNN-LSTM models with built-in explainability (SHAP, GradCAM) for robust, transparent predictions.
2.	LLMs & XAI for Water Sustainability: Seasonal Water Quality Prediction (arXiv)	2025	Focuses on text-based seasonal prediction, limited direct linkage to remote sensing, lacks ground validation.	Fuse remote sensing input with XAI-driven model; validate predictions using ground truth from open datasets.
3.	A Multi-Step Approach for Water Quality Parameter Estimation Using Deep Learning (MDPI)	2022	Focus on optical parameters only and single-site biases; lacks multi-temporal and multi-site evaluation.	Expand to multi-site, multi-sensor, multi-temporal datasets including both optical and non-optical variables.
4.	Monitoring Water Quality Parameters Using Multi-Sensor Satellite Imagery and ML (TandF)	2025	No systematic benchmarking across architectures; model performance varies widely between sensors and sites.	Responsible benchmarking of RF, CNN, CNN-LSTM, and hybrid models across multiple sensors/sites, with transfer learning.

Research Gaps

- Traditional water quality monitoring methods used by regulatory agencies rely on manual sampling and laboratory analysis, which are accurate but slow, expensive, and spatially sparse.
- Many existing remote sensing–based water quality studies are limited to single rivers or localized regions, resulting in poor scalability and limited generalization.
- Several machine learning approaches focus on either physicochemical data or satellite imagery alone, without effectively integrating ground truth measurements with remote sensing features.
- Deep learning–based water quality models are often treated as black boxes, offering little interpretability, which limits their adoption by policymakers and environmental authorities.
- Most existing models prioritize overall accuracy, while safety-critical metrics such as recall for polluted water are not adequately emphasized.
- There is limited research demonstrating explainable machine learning frameworks that can justify pollution predictions using physically meaningful satellite features.
- Few studies explicitly design water quality monitoring systems to support national-scale deployment and SDG-aligned environmental governance.

Methodology

The proposed system integrates satellite remote sensing, ground-truth water quality data, and machine learning with explainability to enable automated water quality monitoring.

Data

National CPCB water quality data (DO, BOD, pH, conductivity) is used as ground truth, along with Sentinel-2 multispectral satellite imagery for selected river stretches.

Acquisition:

Preprocessing:

CPCB records are cleaned and labeled into a binary classification problem (Moderate vs Poor). Satellite images undergo cloud masking, normalization, and water-pixel extraction.

Feature

Key Sentinel-2 bands (Blue, Green, Red, NIR, SWIR) and derived indices (NDWI, NDTI) are extracted due to their sensitivity to turbidity and pollution.

Extraction:

Methodology

Modeling:

Random Forest (RF): Baseline model for tabular spectral features; robust and interpretable.

CNN: Learns spatial–spectral patterns directly from satellite image patches.

CNN-LSTM: Extends CNN by modeling temporal trends across multi-date observations.

Evaluation:

Models are evaluated using precision, recall, F1-score, and confusion matrix, with emphasis on high recall for polluted (Poor) water to ensure safety.

Explainability:

SHAP explains RF predictions, while Grad-CAM (planned) visualizes influential regions in CNN/CNN-LSTM outputs.

Validation:

Cross-region and multi-temporal validation is performed to assess scalability and generalization.



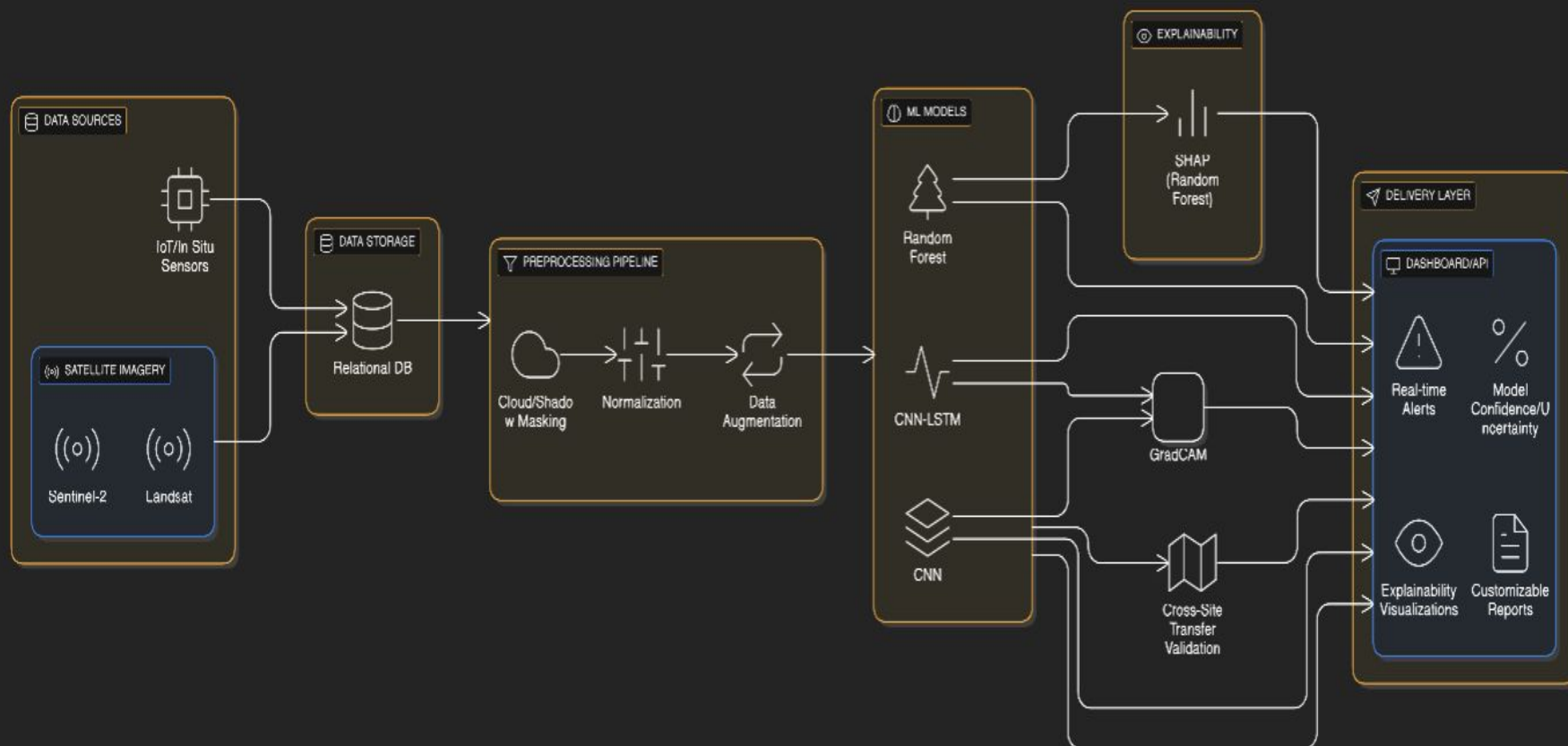
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Proposed System

The proposed system is an end-to-end, automated machine learning pipeline that fuses multispectral satellite imagery and situ/IOT sensor data to classify and predict water quality across diverse water bodies. The system leverages a hybrid model architecture—combining Random Forest, Convolutional Neural Networks (CNN), and CNN-LSTM—to capture spatial, spectral, and temporal signals. Model predictions are made interpretable and actionable via integrated explainability modules (SHAP for classical ML, GradCAM for deep learning). The framework supports rigorous cross-validation across multiple regions and sensors to ensure real-world transferability, and is designed for usability by water resource managers and policymakers.

Architecture Diagram





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Key Features

- **Hybrid Architecture:** Fuses Random Forest, CNN, and CNN-LSTM for robust spatial-temporal modeling and improved accuracy over single-model approaches.
- **Multi-Source Data:** Supports input from both remote sensing (Sentinel, Landsat) and sensor/in situ measurements, enabling holistic assessment.
- **Explainable AI:** Embeds SHAP and GradCAM for transparent model outputs; users can see which features or imagery bands drive key predictions.
- **Cross-Site Transferability:** Benchmarks model performance across different geographic regions, river/lake types, and satellite sensors.
- **Modular & Scalable:** Easily extendable to new water bodies or additional sensors, with automated preprocessing and retraining workflows.
- **Open Access:** Designed for deployment with open, freely available datasets.



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Results / Expected Outcome

- **Increased Accuracy:** Significant gains in water quality classification metrics (e.g., F1-score, accuracy) when compared to single-method baselines and classical ML approaches.
- **Interpretability:** End-users receive actionable insights, with clear explanations of what features (e.g., specific bands or dates) influenced each water quality prediction.
- **Robust Generalization:** Model maintains performance across new regions and sensor types, demonstrating cross-site robustness unprecedented in prior work.
- **Scalable Deployment:** System is modular and lightweight, supporting integration into government or NGO open data dashboards to monitor routine water quality at scale.
- **SDG 6 Impact:** Supports rapid identification of pollution events and trends, informing better water resource management and policy interventions.

Conclusion

This project presents a practical and innovative hybrid machine learning framework for automated water quality monitoring, addressing key gaps in accuracy, interpretability, and regional robustness identified in recent literature. By seamlessly combining classical and deep learning with explainable AI, and validating across multiple sites and sensors, the proposed system advances the state-of-the-art in SDG 6 monitoring. The approach promises scalable, reliable insights for water quality management—supporting regulatory agencies, environmental NGOs, and the wider SDG community in the pursuit of clean and safe water for all.